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Entropy-based regularized regression for advanced distribution of relaxation times deconvolution

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HIGHLIGHTS

- Entropic regularization is able to precisely remove meaningless peaks in DRT spectra.
- Entropic regularization can be recast in a Bayesian framework.
- L-curve method can be used to select the regularization parameter for entropic DRT.
- Entropic regularization is more accurate to recover multidimensional DRTs.

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ABSTRACT

Electrochemical impedance spectroscopy (EIS) coupled with distribution of relaxation times (DRT) analysis is a robust framework for characterizing electrochemical systems. However, DRT deconvolution is often plagued by spurious peaks, hindering accurate process identification and quantitative parameter estimation. To overcome this critical limitation, we rigorously evaluate entropy-based regularization within the standard regularized regression framework for DRT deconvolution. Building upon initial investigations, this study provides a detailed and quantitative evaluation of entropy-regularized DRT using a comprehensive suite of simulated impedance spectra, encompassing various impedance models and noise levels, and experimental data from lithium-ion batteries, fuel cells, and solar cells. Our results demonstrate that entropy regularization consistently outperforms conventional ridge regression, achieving significantly improved accuracy in DRT recovery and effectively eliminating spurious peaks. Furthermore, we demonstrate the novel capabilities of the entropy method for Bayesian DRT uncertainty quantification through sampling and for the simultaneous analysis of multiple impedance spectra. These findings establish entropy-based deconvolution as a superior and versatile technique for robust and insightful electrochemical system characterization, advancing the reliability of DRT analysis in electrochemistry.

1. Introduction

Electrochemical impedance spectroscopy (EIS) is a key technique for examining the complex dynamics of electrochemical systems, encompassing batteries, fuel cells, and sensors across a broad spectrum of timescales [1–3]. By analyzing system response across varying frequencies, EIS provides insights into electrochemical mechanisms, device performance, and degradation pathways [4]. However, the complexity of EIS data often presents a significant analytical challenge as interpreting intricate impedance spectra to extract physical parameters and identify underlying processes remains difficult. To address this, the distribution of relaxation times (DRT) method has emerged as an effective alternative to conventional equivalent circuit models [5, 6]. DRT decomposes the impedance response into a distribution of time constants, enabling the association of timescales with specific electrochemical processes without relying on *a priori* circuit assumptions. Mathematically, the DRT-based impedance, $Z_{\mathrm{DRT}}(f)$, at a given frequency, f, is expressed as [7]

$$Z_{\mathrm{DRT}}(f) = R_\infty + \int_{-\infty}^{+\infty} \frac{\gamma(\log \tau)}{1 + i2\pi f \tau} \, \mathrm{d} \log \tau \tag{1}$$

where R_∞ represents the ohmic resistance, $\gamma(\log \tau)$ is the DRT function, and τ denotes the relaxation time constant.[1]

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[1] The polarization resistance, R_{pol}, is given by $R_{\mathrm{pol}} = Z_{\mathrm{DRT}}(0) - Z_{\mathrm{DRT}}(f \to +\infty) = \int_{-\infty}^{+\infty} \gamma(\log \tau) \, \mathrm{d} \log \tau$.

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Recovering the DRT, $\gamma(\log\tau)$, from experimental EIS data involves solving an ill-posed inverse problem, necessitating methods that ensure solution stability and prevent overfitting. While various approaches have emerged, including Fourier transform [8], neural networks [9], regularized regression [10], and Gaussian processes [11], regularized regression has gained widespread adoption. This method estimates the DRT by solving the following optimization problem [12]:

$$\hat{\mathbf{x}} = \underset{\mathbf{x}>\mathbf{0}}{\arg\min}\left(\|\mathbf{A}\mathbf{x}-\mathbf{Z}_{\exp}\|^2 + \lambda\mathcal{P}(\mathbf{x})\right) \tag{2}$$

where $\mathbf{x} = \begin{pmatrix} R_\infty \\ \boldsymbol{\gamma} \end{pmatrix}$ is the $M+1$ discretized DRT vector (M is the number of EIS frequencies) with $\boldsymbol{\gamma} = (\gamma(\log\tau_1), \gamma(\log\tau_2), \ldots, \gamma(\log\tau_N))^\top$ being the N-dimensional DRT vector, $\mathbf{A}$ is the discretization matrix, $\hat{\mathbf{x}}$ is the estimate of $\mathbf{x}$, $\mathbf{Z}_{\exp}$ is the experimental impedance data vector, and $\mathcal{P}(\mathbf{x})$ is the penalty function weighted by the scalar λ. The penalty term is essential for enforcing smoothness and mitigating the formation of spurious peaks, which can hinder the identification of genuine electrochemical features [9,10], as seen for instance in the case of lithium-sulfur batteries [13]. Common penalty functions include L2-norm (ridge regression, RR), L1-norm (lasso regression), and elastic net regularization [14,15].

An alternative regularization approach uses Shannon entropy as the penalty function [16,17], building on early DRT investigations by Hörlin [18,19] and van der Noot [20]. Our implementation utilizes a non-normalized, Bayesian interpretation of entropy, with the penalty defined as

$$\mathcal{P}(\mathbf{x}) = \text{entropy}(\mathbf{x}) = -\sum_{k=1}^{N+1} x_k \log x_k \tag{3}$$

In the Bayesian framework, entropy maximization serves as a prior that favors smoother, more distributed solutions [16]. This approach, implemented through (3), constrains the solution space by promoting simpler (more entropic) DRT spectra, thereby reducing noise amplification and preventing the formation of artificial peaks, favoring distributed time constants rather than discrete relaxation times. While entropy regularization has successfully addressed various ill-posed problems [21–26], its application to electrochemical DRT analysis remains largely unexplored, particularly within a Bayesian regularized regression framework.

This article evaluates entropy-based DRT deconvolution through quantitative comparison with conventional RR. Our analysis, encompassing both simulated and experimental impedance data, demonstrates the superior performance of entropy regularization, yielding more accurate DRT recovery and effective suppression of spurious peaks. This enhanced performance persists across overlapping processes and elevated noise levels in impedance spectra. Quantitatively, entropy regularization reduces DRT recovery errors by approximately 50% compared to RR while maintaining high impedance fitting accuracy. Moreover, this study advances DRT analysis through two key methodological innovations. The first is a robust framework for uncertainty quantification, achieved by applying Monte Carlo sampling to the entropic regression DRT method within a Bayesian framework. The second is a versatile multi-spectrum processing capability that enables systematic tracking of electrochemical dynamics across varying experimental conditions. We validate these advances through extensive testing on diverse electrochemical systems, including lithium-ion batteries, fuel cells, and solar cells, and demonstrate that entropy-based deconvolution significantly enhances both the reliability and interpretability of DRT analysis in electrochemistry and related fields.

2. Methods

2.1. Theory

To frame entropic regularization mathematically, we leverage the Bayesian DRT approach, and write [12]

$$p(\mathbf{Z}_{\exp})\, p(\mathbf{x}|\mathbf{Z}_{\exp}) = p(\mathbf{x})\, p(\mathbf{Z}_{\exp}|\mathbf{x}) \tag{4}$$

where $p(\cdot)$ is a probability distribution function, $p(\mathbf{Z}_{\exp})$ is the experimental evidence, and $p(\mathbf{Z}_{\exp}|\mathbf{x})$ is the likelihood. The likelihood is given by

$$p(\mathbf{Z}_{\exp}|\mathbf{x}) \propto \exp\left(-\frac{1}{2\sigma^2}\|\mathbf{A}\mathbf{x}-\mathbf{Z}_{\exp}\|^2\right) \tag{5}$$

where σ is the standard deviation of the experimental noise.

The prior, $p(\mathbf{x})$, encodes the known knowledge on $\mathbf{x}$ as [12]

$$p(\mathbf{x}) \propto \exp\left(-\frac{\lambda}{2\sigma^2}\mathcal{P}(\mathbf{x})\right) \tag{6}$$

Consequently, the posterior, $p(\mathbf{x}|\mathbf{Z}_{\exp})$, is obtained by combining (4), (5), and (6) as

$$p(\mathbf{x}|\mathbf{Z}_{\exp}) = \frac{p(\mathbf{x})\, p(\mathbf{Z}_{\exp}|\mathbf{x})}{p(\mathbf{Z}_{\exp})} \propto \exp\left(-\frac{1}{2\sigma^2}\left(\|\mathbf{A}\mathbf{x}-\mathbf{Z}_{\exp}\|^2 + \lambda\mathcal{P}(\mathbf{x})\right)\right) \tag{7}$$

A common approach in DRT analysis begins with selecting the prior function $\mathcal{P}(\mathbf{x}) = \mathbf{x}^\top\mathbf{L}^\top\mathbf{L}\mathbf{x}$, where the choice of matrix $\mathbf{L}$ fundamentally shapes the probability distribution properties. Three key cases arise. First, when $\mathbf{L} = \mathbf{I}$ (zeroth-order derivative), the term $\mathbf{L}^\top\mathbf{L} = \mathbf{I}^\top\mathbf{I} = \mathbf{I}$ is positive definite, yielding a proper probability distribution function. Second and third, when $\mathbf{L}$ represents first-order ($\mathbf{L}_1$) or second-order ($\mathbf{L}_2$) differentiation matrices, $\mathbf{L}^\top\mathbf{L}$ becomes positive semi-definite. All three formulations allow a solution of (2) [14], which is obtained by maximizing the posterior probability $p(\mathbf{x}|\mathbf{Z}\exp)$ with respect to $\mathbf{x}$.

To recover the entropic formulation with (3), we can set the prior as

$$p(\mathbf{x}) = \prod_{n=1}^{N+1} \frac{1}{\zeta\left(\frac{\lambda}{2\sigma^2}\right)} \exp\left(-\frac{\lambda}{2\sigma^2} x_n \log x_n\right) \tag{8}$$

where the partition function, $\zeta\left(\frac{\lambda}{2\sigma^2}\right)$, is defined as

$$\zeta\left(\frac{\lambda}{2\sigma^2}\right) = \int_0^{+\infty} \exp\left(-\frac{\lambda}{2\sigma^2} x \log x\right)\, \mathrm{d}x \tag{9}$$

In the absence of strong evidence in the impedance data for highly specific relaxation processes, the entropy penalty guides the solution towards a more entropic DRT that is more uniformly distributed across the timescales. As explained in the introduction (Section 1), this approach effectively mitigates the propensity for noise in ill-posed inverse problems to be amplified into spurious, localized features in the recovered DRT spectrum.

We now extend the entropy-based regularization, initially derived for a single EIS experiment, to handle a set of measurements conducted under varying experimental conditions. Consider that we perform K EIS measurements while systematically varying a scalar experimental state variable $\boldsymbol{\psi} = (\psi_1, \psi_2, \ldots, \psi_K)^\top$; examples of such state variables include the state of charge for batteries (as discussed in Section 3.2.1) [9]. In this multidimensional context, we generalize the optimization problem in (2) to

$$\hat{\mathbf{X}} = \underset{\mathbf{x}>\mathbf{0}}{\arg\min}\left(\sum_{k=1}^{K}\|\mathbf{A}_k\mathbf{x}_k - \mathbf{Z}_{\exp,k}\|^2 - \lambda\sum_{k=1}^{K}\text{entropy}(\mathbf{x}_k)\right) \tag{10}$$

where $\hat{\mathbf{X}} = (\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2, \ldots, \hat{\mathbf{x}}_K)$ represents the DRT matrix obtained from each individual experiment k with $k = 1, 2, \ldots, K$ and $\mathbf{x}_k$, $\mathbf{A}_k$, and $\mathbf{Z}_{\exp,k}$ being the discretized DRT vector, the discretization matrix, and the vector of experimental impedances, respectively.

To further refine the model, and endow the DRT with smoothness with respect to the experimental state variable $\boldsymbol{\psi}$, we augment the penalty term in (10), leading to following optimization problem

$$\hat{\mathbf{x}} = \underset{\mathbf{x}>\mathbf{0}}{\arg\min}\left(\sum_{k=1}^{K}\|\mathbf{A}_k\mathbf{x}_k - \mathbf{Z}_{\exp,k}\|^2 - \lambda\sum_{k=1}^{K}\text{entropy}(\mathbf{x}_k) + \lambda_\tau\sum_{k=1}^{K}\|\mathbf{L}_\tau\mathbf{x}_k\|^2 + \lambda_\psi\sum_{k=1}^{K}\|\mathbf{L}_\psi\mathbf{x}_k\|^2\right) \tag{11}$$

where $\mathbf{L}_\tau$ and $\mathbf{L}_\psi$ denote the second-order differentiation matrices with respect to $\log \tau$ and ψ, respectively, and λ_τ and λ_ψ are the corresponding regularization parameters that control the smoothness imposed with respect to the relaxation timescale and experimental state. While this work focuses on a single (scalar) experimental state variable, the framework can be readily extended to multiple experimental state variables, which can be represented as vectors [9]. Furthermore, it is important to highlight that Eq. (11) implies experimental states function independently without interconnected relationships [27,28].

2.2. Regularization parameter selection and performance metrics

To optimize the entropy-based DRT deconvolution, we evaluate four methods for selecting the regularization parameter λ: the L-curve (LC), discrepancy principle[2] (DP), stability (SB), and second derivative (SD) methods. Each method determines λ by minimizing a specific score function [16,29]

$$\lambda_{\mathrm{LC}} = \arg\min_{\lambda>0} \frac{\frac{\mathrm{d}^2\kappa(\lambda)}{\mathrm{d}\lambda^2}\frac{\mathrm{d}\theta(\lambda)}{\mathrm{d}\lambda} - \frac{\mathrm{d}\kappa(\lambda)}{\mathrm{d}\lambda}\frac{\mathrm{d}^2\theta(\lambda)}{\mathrm{d}\lambda^2}}{\left(\left(\frac{\mathrm{d}\kappa(\lambda)}{\mathrm{d}\lambda}\right)^2 + \left(\frac{\mathrm{d}\theta(\lambda)}{\mathrm{d}\lambda}\right)^2\right)^{\frac{3}{2}}} \tag{12a}$$

$$\lambda_{\mathrm{DP}} = \arg\min_{\lambda>0} \kappa(\lambda) \tag{12b}$$

$$\lambda_{\mathrm{SB}} = \arg\min_{\lambda>0} \frac{\mathrm{d}\,\|\hat{\mathbf{x}}(\lambda)\|}{\mathrm{d}\lambda} \tag{12c}$$

$$\lambda_{\mathrm{SD}} = \arg\min_{\lambda>0} \frac{\mathrm{d}^2\,\|\hat{\mathbf{x}}(\lambda)\|}{\mathrm{d}\lambda^2} \tag{12d}$$

where $\kappa(\lambda) = \|\mathbf{A}\hat{\mathbf{x}}(\lambda) - \mathbf{Z}_{\mathrm{exp}}\|^2$ is the squared impedance residual, and $\theta(\lambda) = \lambda\,\mathrm{entropy}(\hat{\mathbf{x}}(\lambda))$ is the entropic penalty. Here, $\hat{\mathbf{x}}(\lambda)$ represents the DRT vector obtained by solving (2) for a given λ.

To assess the accuracy of our deconvolution approach, we employ normalized residuals for both the DRT (ε_γ) and impedance spectra (ε_Z), given by

$$\varepsilon_\gamma(\lambda) = \frac{\|\boldsymbol{\gamma}_{\mathrm{exact}} - \hat{\boldsymbol{\gamma}}(\lambda)\|^2}{\|\boldsymbol{\gamma}_{\mathrm{exact}}\|^2} \tag{13a}$$

$$\varepsilon_Z(\lambda) = \frac{\|\mathbf{Z}_{\mathrm{exact}} - \hat{\mathbf{Z}}(\lambda)\|^2}{\|\mathbf{Z}_{\mathrm{exact}}\|^2} \tag{13b}$$

where $\boldsymbol{\gamma}_{\mathrm{exact}}$ and $\mathbf{Z}_{\mathrm{exact}}$ denote the analytically known DRT and impedance, respectively (detailed in Section 2.3.3).

2.3. Implementation methods

2.3.1. DRT optimization

The integral in (1) was discretized using piecewise-linear functions and standard log-equispaced nodal timescales, spanning from 10^{-6} s to 10^2 s, with a density of ten points per decade [10]. DRT recovery was performed using the MOSEK solver [30] within the CVXPY framework [31]. This involved minimizing the objective function in (2) for the given penalty. The regularization parameter λ was optimized by minimizing the DRT residual $\varepsilon_\gamma(\lambda)$ (defined in ((13a)) across the λ range of 10^{-6}–10^2. For the two-dimensional synthetic datasets (detailed in Section 3.1.3), the regularization parameters λ, λ_τ, and λ_ψ were simultaneously optimized by minimizing ((13a)) over $[10^{-8}, 10^4] \times [10^{-8}, 10^4] \times [10^{-8}, 10^4]$. We set $\lambda = \lambda_\tau = \lambda_\psi = 10^{-3}$ for real data (Section 3.2).

[2] It is worth noting that the discrepancy principle corresponds to evidence maximization.

2.3.2. Uncertainty quantification through sampling

To characterize the uncertainty in the DRT recovery, the posterior probability distribution $p(\mathbf{x}|\mathbf{Z}_{\mathrm{exp}})$ (introduced in Section 2.1) was sampled using the No-U-Turn sampler (NUTS) algorithm. NUTS represents an efficient Markov Chain Monte Carlo algorithm based on Hamiltonian Monte Carlo chosen specifically for its effectiveness in sampling complex, high-dimensional posterior distributions. The algorithm enhances standard HMC by adaptively determining the trajectory length, avoiding inefficient "U-turns" in parameter space, thus improving sampling speed and exploration [32].

The implementation used the pymc library [32,33] and was initialized with the DRT solution obtained from either the entropy method or RR, as described in Section 2.3.1. Four independent chains with 10,000 draws each were used to ensure convergence and robust estimation of posterior statistics, balancing computational cost with sufficient sample size for reliable uncertainty quantification. The acceptance rate was set to 95%, and the results are presented graphically, displaying the median (solid line), maximum a posteriori (MAP) estimate (dots), and 95% credible intervals (shaded areas).

2.3.3. Generation of synthetic experiments

Synthetic impedance data was generated by adding white Gaussian noise to exact impedance spectra. Specifically, 100 artificial EIS spectra were created for each impedance model by adding random noise $\epsilon \sim \sigma_n^{\mathrm{exp}}\mathcal{N}(0,1)$, where σ_n^{exp} was set to 0.5 Ω or 1.0 Ω. This noise was added to the exact impedance functions $Z_{\mathrm{exact}}(f)$ [10]. The chosen reference impedance models were the single ZARC, double ZARC, Gerischer, and fractal configurations [9,10]. Frequency points were logarithmically distributed between 10 mHz and 1 MHz, with a density of ten frequencies per decade. Table S1 of the Supplementary Information (SI) lists the specific circuit parameter values used for each model.

3. Results

Our investigation followed a two-phase strategy. First, we systematically validated the entropy-based DRT method using synthetic EIS data to assess its performance under controlled conditions (Section 3.1). Then, we demonstrated its practical utility by analyzing experimental impedance spectra from lithium-ion batteries, protonic ceramic fuel cells, and perovskite solar cells (Section 3.2). This approach enabled rigorous validation before tackling real-world electrochemical characterization challenges.

3.1. Synthetic experiments

3.1.1. Preliminary validation

We commenced our analysis by examining synthetic data generated using the single ZARC model with a noise level of $\sigma_n^{\mathrm{exp}} = 0.5$ Ω (circuit parameter values are detailed in Table S1). Fig. 1(a) presents the normalized DRT residuals, ε_γ, for each regularization method across a range of regularization parameter, λ, from 10^{-6} to 10^2. Notably, the entropy method yielded the lowest ε_γ value, achieving an error magnitude at least one order of magnitude smaller than those observed with regularization using each differentiation matrix. This superior performance stems from the entropy method's ability to accurately recover the DRT without introducing spurious ridges at the main peak's edges, as visually confirmed in Fig. 1(c) and (e), as well as Figure S1 (a) and (c). In terms of impedance recovery, all regularization methods performed comparably well, as indicated by the normalized impedance residuals in Figure S2 (a). To further substantiate these initial findings, we extended our analysis to 100 synthetic spectra. The additional results corroborated the initial observations, demonstrating that the entropy method consistently produced the lowest DRT residuals, as shown in the violin plots in Figure S3 (a) and the mean values summarized in Table 1, which were 50% smaller than the values obtained with the other regularization methods.

Table 1
For 100 synthetic spectra and each impedance model, normalized DRT (ε_γ) and impedance (ε_Z) residuals obtained with each regularization method. The lowest residual for a given impedance model is highlighted in red.

$\sigma_n^{\exp} = 0.5\ \Omega$								
Impedance model	**I**		$\mathbf{L}_1$		$\mathbf{L}_2$		Entropy	
	$\varepsilon_\gamma \times 10^{-2}$	$\varepsilon_Z \times 10^{-3}$	$\varepsilon_\gamma \times 10^{-2}$	$\varepsilon_Z \times 10^{-3}$	$\varepsilon_\gamma \times 10^{-2}$	$\varepsilon_Z \times 10^{-3}$	$\varepsilon_\gamma \times 10^{-2}$	$\varepsilon_Z \times 10^{-3}$
1×ZARC	2.42 ± 1.15	0.33 ± 0.04	2.21 ± 1.05	0.33 ± 0.04	2.10 ± 1.03	0.33 ± 0.04	0.80 ± 0.58	0.34 ± 0.04
Distant 2×ZARC	2.64 ± 1.05	0.17 ± 0.02	2.48 ± 1.02	0.17 ± 0.02	2.43 ± 1.04	0.17 ± 0.02	0.92 ± 0.52	0.18 ± 0.02
Separated 2×ZARC	2.21 ± 0.84	0.08 ± 0.01	2.07 ± 0.82	0.08 ± 0.01	2.00 ± 0.83	0.08 ± 0.01	0.76 ± 0.40	0.09 ± 0.01
Overlapping 2×ZARC	1.90 ± 0.88	0.12 ± 0.02	1.73 ± 0.84	0.12 ± 0.02	1.64 ± 0.82	0.12 ± 0.02	1.01 ± 0.65	0.13 ± 0.02
Gerischer	35.6 ± 0.85	0.32 ± 0.04	34.3 ± 0.67	0.31 ± 0.04	34.0 ± 0.76	0.31 ± 0.04	27.7 ± 0.67	0.37 ± 0.05
Fractal	52.0 ± 0.41	1.38 ± 0.22	52.6 ± 0.49	0.58 ± 0.08	52.6 ± 0.54	0.55 ± 0.08	40.8 ± 0.42	1.68 ± 0.18
$\sigma_n^{\exp} = 1.0\ \Omega$								
Impedance model	**I**		$\mathbf{L}_1$		$\mathbf{L}_2$		Entropy	
	$\varepsilon_\gamma \times 10^{-2}$	$\varepsilon_Z \times 10^{-3}$	$\varepsilon_\gamma \times 10^{-2}$	$\varepsilon_Z \times 10^{-3}$	$\varepsilon_\gamma \times 10^{-2}$	$\varepsilon_Z \times 10^{-3}$	$\varepsilon_\gamma \times 10^{-2}$	$\varepsilon_Z \times 10^{-3}$
1×ZARC	3.81 ± 1.79	1.33 ± 0.16	3.44 ± 1.64	1.32 ± 0.16	3.27 ± 1.67	1.32 ± 0.16	1.46 ± 1.04	1.36 ± 0.17
Distant 2×ZARC	4.09 ± 1.52	0.69 ± 0.09	3.78 ± 1.38	0.68 ± 0.09	3.67 ± 1.42	0.68 ± 0.09	1.69 ± 0.96	0.70 ± 0.09
Separated 2×ZARC	3.50 ± 1.26	0.33 ± 0.04	3.29 ± 1.20	0.33 ± 0.04	3.20 ± 1.21	0.33 ± 0.04	1.40 ± 0.72	0.34 ± 0.04
Overlapping 2×ZARC	3.14 ± 1.34	0.50 ± 0.06	2.92 ± 1.30	0.50 ± 0.06	2.81 ± 1.32	0.50 ± 0.06	1.89 ± 1.16	0.51 ± 0.06
Gerischer	36.2 ± 1.46	1.22 ± 0.16	35.0 ± 1.26	1.19 ± 0.15	34.6 ± 1.42	1.19 ± 0.15	27.9 ± 1.28	1.27 ± 0.17
Fractal	52.0 ± 0.82	2.33 ± 0.48	52.7 ± 0.96	1.52 ± 0.20	52.6 ± 1.05	1.48 ± 0.19	40.8 ± 0.83	2.60 ± 0.39

Next, we shifted our focus to the distant 2×ZARC model (parameter values in Table S1). The recovered DRTs are depicted in the right panel of Fig. 1 and Figure S1, while the impedances regressed using RR and the entropy method are presented in the left and right panels of Fig. 2, respectively. The exact DRT (Fig. 1(d) and (f)) and impedance (Figure S2 (b)) for this model were accurately reconstructed by both methods. This finding was reinforced by analyzing 100 artificial spectra (Figure S3 (b)) for each of the three 2×ZARC models considered in this study (Table 1). Like for the single ZARC model, the DRT error obtained with the entropy method was significantly reduced compared to the other regularization methods. Furthermore, we employed the NUTS sampler (Section 2.3.2) to sample the DRTs recovered via RR (Fig. 1(d)) and the entropy method (Fig. 1(f)). The resulting sampled DRTs are displayed in Fig. 3(a) and (b). Crucially, the median DRT obtained with the entropy method exhibited closer agreement with the exact DRT and lacked spurious peaks, unlike the mean DRT derived from RR.

Moreover, the entropy method demonstrated accurate recovery of the discontinuous DRTs associated with the Gerischer and fractal models. This was supported by the normalized DRT and impedance residuals detailed in Table 1. In addition, Fig. 4(a) and (b) illustrate the Gerischer DRTs sampled from the DRTs recovered using RR and the entropy method, respectively. Analogous to the distant 2×ZARC model, the median entropy-based DRT for the Gerischer model was closer to the exact DRT and free from artifacts. It is important to emphasize that the superior performance of the entropy method remained consistent even at a higher experimental noise level, $\sigma_n^{\exp} = 1.0\ \Omega$, as confirmed by our validation at this noise level (Table 1).

3.1.2. Optimizing the regularization parameter

In the preceding preliminary validation (Section 3.1.1), the regularization parameter λ was chosen by minimizing the normalized DRT residual, a strategy feasible because the exact DRT is known for synthetic data. However, this approach is not applicable to experimental data where the true DRT is unknown (Section 3.2). Consequently, we evaluated the effectiveness of the four regularization parameter selection methods detailed in Section 2.2 by comparing the λ values they yielded to the optimal λ determined through minimization of ε_γ. The correlation matrix presented in Fig. 5(a) revealed that the regularization parameters derived from the SD and LC methods exhibited the strongest correlation with the optimal values. Furthermore, analysis of the absolute errors and mean squared errors, summarized in Table 2, underscored that the LC method produced λ values closest to the optimal, followed by the DP method, which were more than one order of magnitude larger. This trend was further substantiated by the violin plots of the normalized DRT residuals (in log scale), as displayed in Fig. 5(b).

Table 2
For $N_{\exp} = 100$ synthetic spectra generated with the distant 2× ZARC model ($\sigma_n^{\exp} = 0.5\ \Omega$), absolute error, $\frac{1}{N_{\exp}} \sum_{k=1}^{N_{\exp}} |\lambda_{\mathrm{opt},k} - \lambda_k|$, and mean squared error, $\frac{1}{N_{\exp}} \sum_{k=1}^{N_{\exp}} (\lambda_{\mathrm{opt},k} - \lambda_k)^2$, of the regularization parameter λ for each selection method (Section 2.2). The lowest error within each column is highlighted in red.

Method	$\frac{1}{N_{\exp}} \sum_{k=1}^{N_{\exp}} \lvert\lambda_{\mathrm{opt},k} - \lambda_k\rvert$	$\frac{1}{N_{\exp}} \sum_{k=1}^{N_{\exp}} (\lambda_{\mathrm{opt},k} - \lambda_k)^2$
LC	1.39×10^{-1}	3.35×10^{-2}
DP	1.32×10^{0}	1.35×10^{-1}
SB	4.48×10^{0}	2.20×10^{2}
SD	1.73×10^{0}	4.33×10^{0}

3.1.3. Multiple experiments

To investigate the application of our deconvolution approach to multidimensional impedance datasets, we first generated synthetic two-dimensional impedance data using the distant 2×ZARC model [9]. Nyquist plots of the resulting synthetic experimental impedances are presented in Figure S4 (a) and (b). To recover the DRTs, we applied (11) in two distinct configurations: first, employing only the entropy regularization term (effectively setting $\lambda_\tau = \lambda_\psi = 0$ while retaining λ); and second, incorporating all three regularization parameters (λ, λ_τ, and λ_ψ). The deconvolved DRTs and the corresponding impedance fits are displayed in the lower and upper panels of Figure S4, respectively, contrasting the results obtained with and without the additional differentiation-based regularization terms. Our analysis revealed that the DRTs deconvolved using all three regularization parameters exhibited closer agreement with the exact DRT (Figure S4 (c)) compared to those obtained using only entropy regularization (Figure S4 (d)). Conversely, the impedance fits achieved comparable accuracy in both configurations, closely matching the exact impedance spectra. To better visualize the impact of the experimental conditions on the recovered DRTs, contour plots are presented in Fig. 6. These plots demonstrate that the incorporation of λ_τ and λ_ψ in (11) enabled a smoother representation of the DRT variations as a function of experimental conditions, as evidenced by the smooth contour lines in Fig. 6(b).

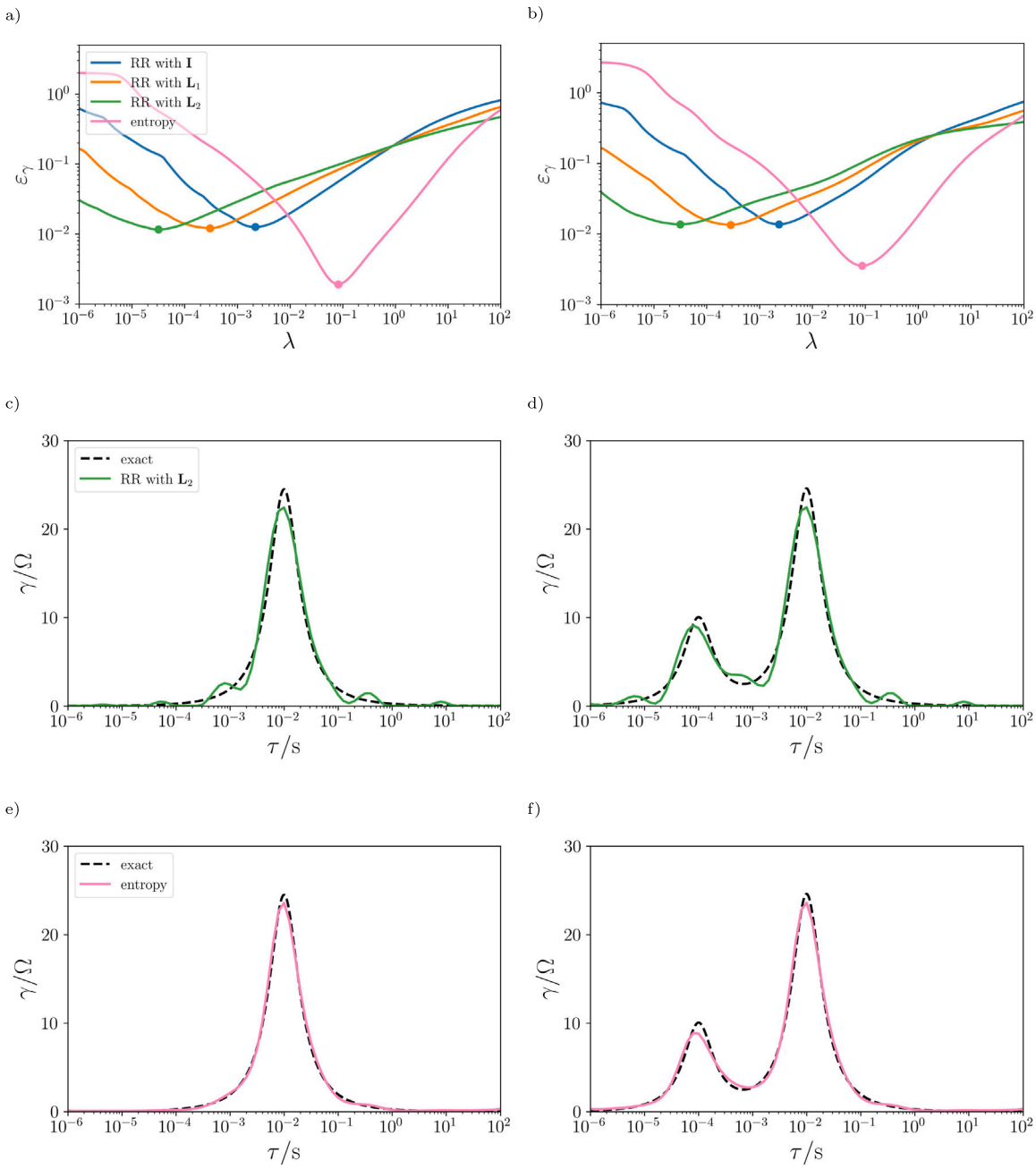

Fig. 1. For one synthetic spectrum generated with the 1×ZARC model ($\sigma_n^{\text{exp}} = 0.5\ \Omega$), (a) normalized DRT residuals (ϵ_γ) for each regularization method, and DRT recovered with (c) RR ($\mathbf{L}_2$) and the (e) entropy method compared to the exact DRT. The corresponding plots for the distant 2×ZARC model are displayed in panels (b), (d), and (f).

3.2. Application to real experimental data

After having established the superior performance of the entropy method in recovering DRTs from synthetic data, particularly in terms of accuracy and artifact suppression (Section 3.1), we demonstrated the feasibility of sampling DRTs obtained with entropy regularization, hence enabling uncertainty quantification. We also recovered the DRT simultaneously from multiple spectra, which is essential for batch processing of real data. We stress however that these conclusions were drawn using artificial and idealized impedance data, which does not always fully mimic experimental data. In this section, we therefore leverage the insights gained from multidimensional DRT analysis (Section 3.1.3) to analyze experimental impedance data acquired from diverse electrochemical systems: two distinct lithium-ion batteries, a protonic ceramic fuel cell, and a perovskite solar cell.

3.2.1. Lithium-ion battery analysis

We initiated our analysis of experimental datasets with a lithium metal battery utilizing a $LiFePO_4$ (LFP) cathode, a 1M $LiPF_6$ electrolyte in ester-based solvents, and a lithium-metal anode [34]. EIS measurements were performed after 30, 60, 90, and 120 charge–discharge cycles at a 5C rate. The Nyquist plots of the measured impedance spectra are displayed in Fig. 7(a). Building upon the findings in Section 3.1.3, we applied multidimensional entropy-based DRT deconvolution with all three regularization parameters ($\lambda = \lambda_\tau = \lambda_\Psi = 10^{-3}$), effectively capturing the influence of cycling on the DRT.

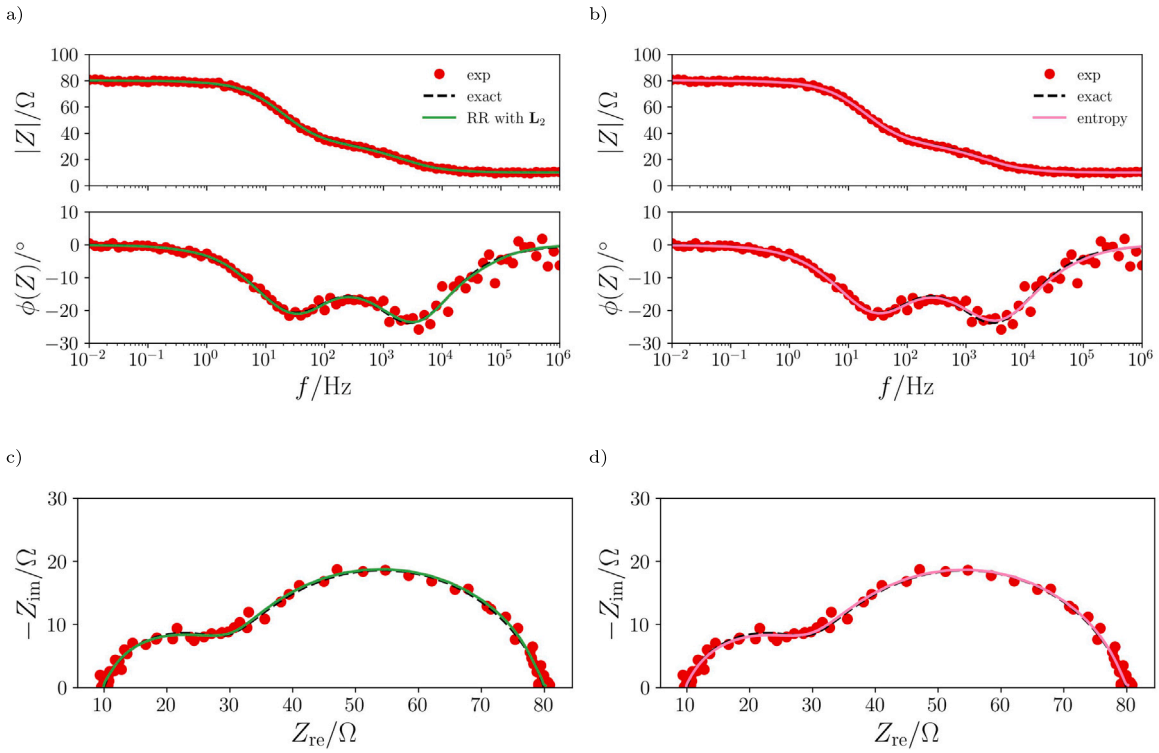

Fig. 2. For one synthetic spectrum generated with the distant 2×ZARC model ($\sigma_n^{\mathrm{exp}} = 0.5\ \Omega$), (a) Bode and (c) Nyquist plots of the impedance recovered with RR ($\mathbf{L}_2$) with the exact and experimental impedances also shown. The corresponding plots for the entropy method are displayed in panels (b) and (d).

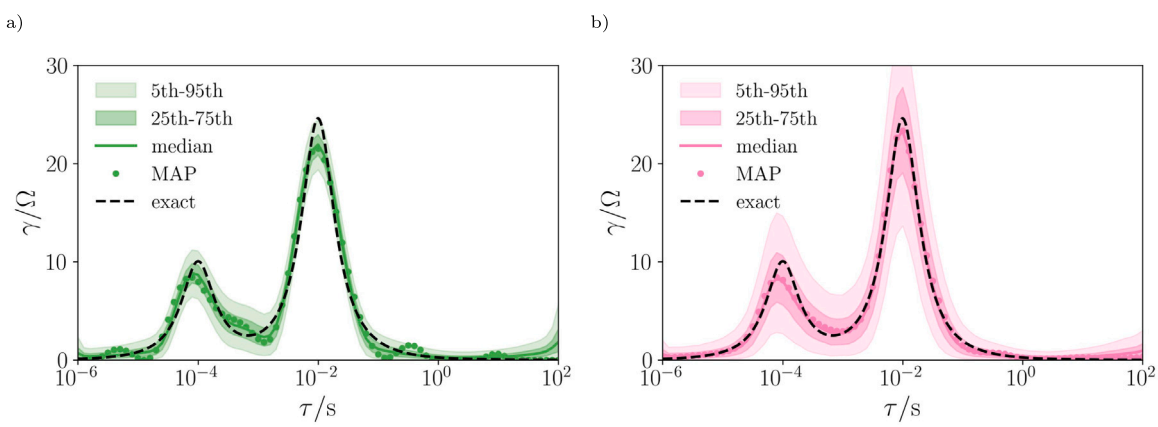

Fig. 3. For the distant 2×ZARC model ($\sigma_n^{\mathrm{exp}} = 0.5\ \Omega$), Bayesian DRTs sampled with (a) RR ($\mathbf{L}_2$) and (b) the entropy method.

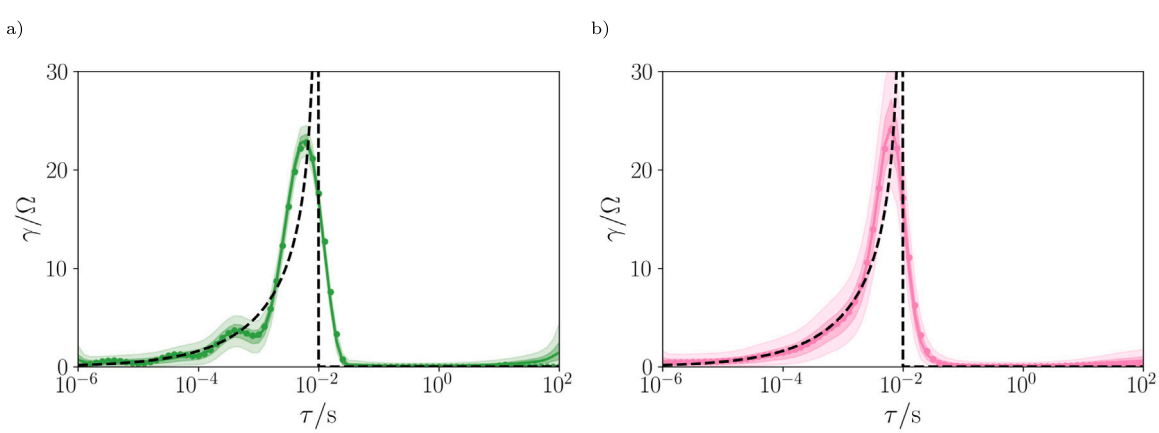

Fig. 4. For the Gerischer model ($\sigma_n^{\mathrm{exp}} = 0.5\ \Omega$), Bayesian DRTs sampled with (a) RR ($\mathbf{L}_2$) and (b) the entropy method.

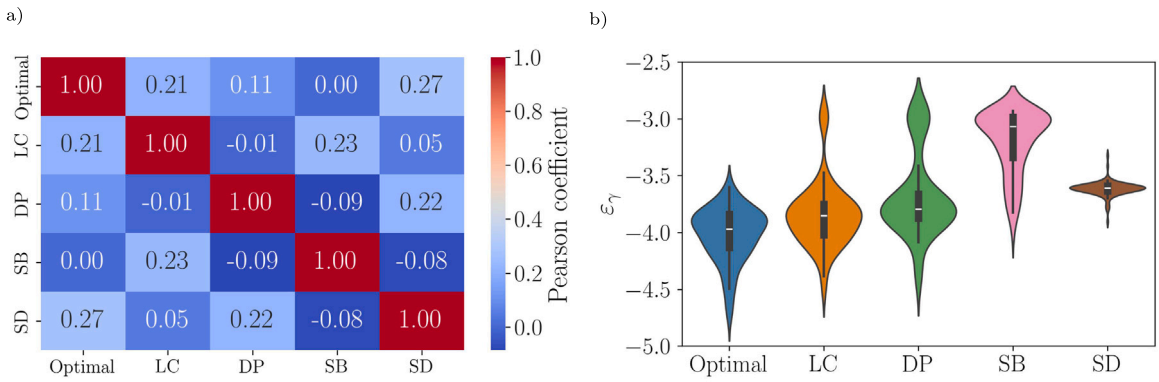

Fig. 5. For 100 synthetic spectra generated with the distant 2×ZARC model ($\sigma_n^{\exp} = 0.5\ \Omega$), (a) correlation matrix illustrating the correlation between optimal λ values and those obtained using each selection method, and (b) violin plots of normalized DRT residuals (ε_γ, log scale) for each selection method (Section 2.2).

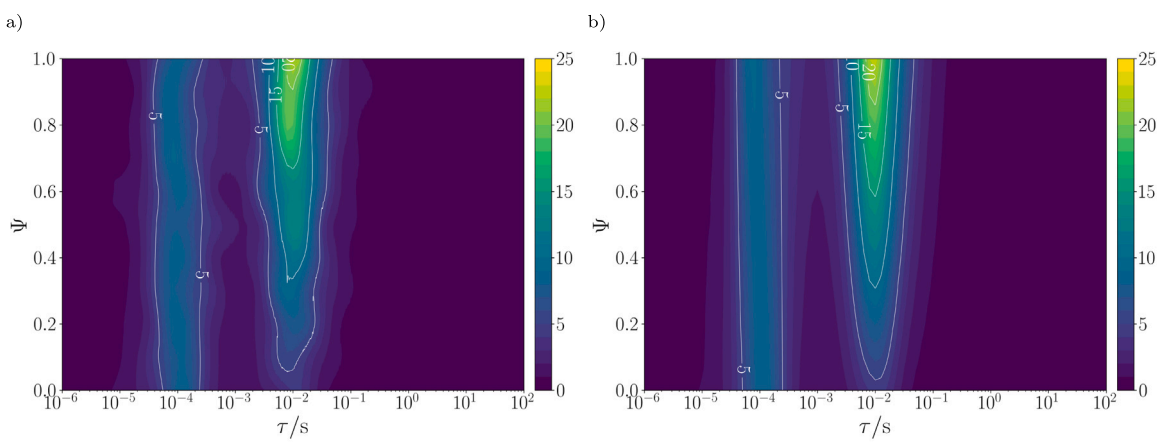

Fig. 6. For 10 synthetic and two-dimensional spectra generated with the distant 2× ZARC model ($\sigma_n^{\exp} = 0.5\ \Omega$), contour plots of the DRTs recovered using the (a) entropy method alone and (b) entropy method combined with differentiation-based regularization (see (11)). The contour lines represent DRT amplitude levels, illustrating the smoothness of the recovered DRT as a function of experimental conditions.

Contour plots of the recovered DRTs, presented in Fig. 7(c), revealed two primary relaxation timescales: one timescale around 0.1 ms, which became more prominent with increasing cycle number, while another timescale at approximately 1.0 s remained relatively stable across cycling. The experimental impedance spectra were accurately fitted by the recovered DRTs, as shown in Fig. 7(a).

Subsequently, we applied the same analytical procedure to impedance data from a Panasonic ML621 battery, measured at various states of charge ranging from 10% to 100% [35] (Fig. 7(b)). The contour plots of the deconvolved DRTs, displayed in Fig. 7(d), indicated a dominant characteristic timescale around 1.0 s. Notably, the amplitude of this peak diminished as the state of charge increased. Furthermore, the experimental impedance spectra were closely regressed by the DRT model, as evidenced in Fig. 7(b).

3.2.2. Protonic ceramic fuel cell analysis

Following the lithium-ion battery analyses, we extended our investigation to impedance data acquired from a protonic ceramic fuel cell. The measurements were conducted on a symmetric cell configuration, employing $Ba_{0.95}La_{0.05}FeO_{3-\delta}$ electrodes and a samarium-doped ceria electrolyte. The cell was operated at 500 °C under varying oxygen partial pressures (pO_2), specifically at 21%, 40%, 60%, 80%, and 100% [36]. Nyquist plots of the experimental impedance spectra are presented in Fig. 8(a). Applying the same DRT deconvolution methodology as used for the battery data (Section 3.2.1), we recovered the DRTs, which are visualized as contour plots in Fig. 8(c). Notably, we observed a consistent trend of decreasing characteristic relaxation timescales with increasing oxygen partial pressure (Fig. 8(c)). This inverse relationship aligns with previously reported findings in the literature [9], suggesting that higher oxygen partial pressures facilitate faster electrochemical processes within the fuel cell. Moreover, the DRT model accurately regressed the experimental impedance spectra, as evidenced in Fig. 8(a).

3.2.3. Perovskite solar cell analysis

Finally, we applied our entropy-based DRT method to analyze impedance spectra from a perovskite solar cell. This device comprised a fluorine-doped tin oxide substrate, an electron-transport layer of SnO_2, and a transport-hole layer of Spiro-OMeTAD. EIS measurements were performed at various operational times to assess device stability (Fig. 8(b)) [9]. The contour plots of the deconvolved DRTs, displayed in Fig. 8 (d), revealed the presence of several distinct characteristic timescales within the solar cell. These timescales likely correspond to different physical and chemical processes occurring within the device. Furthermore, the DRT model accurately regressed the experimental impedance spectra, as evidenced in Fig. 8(b).

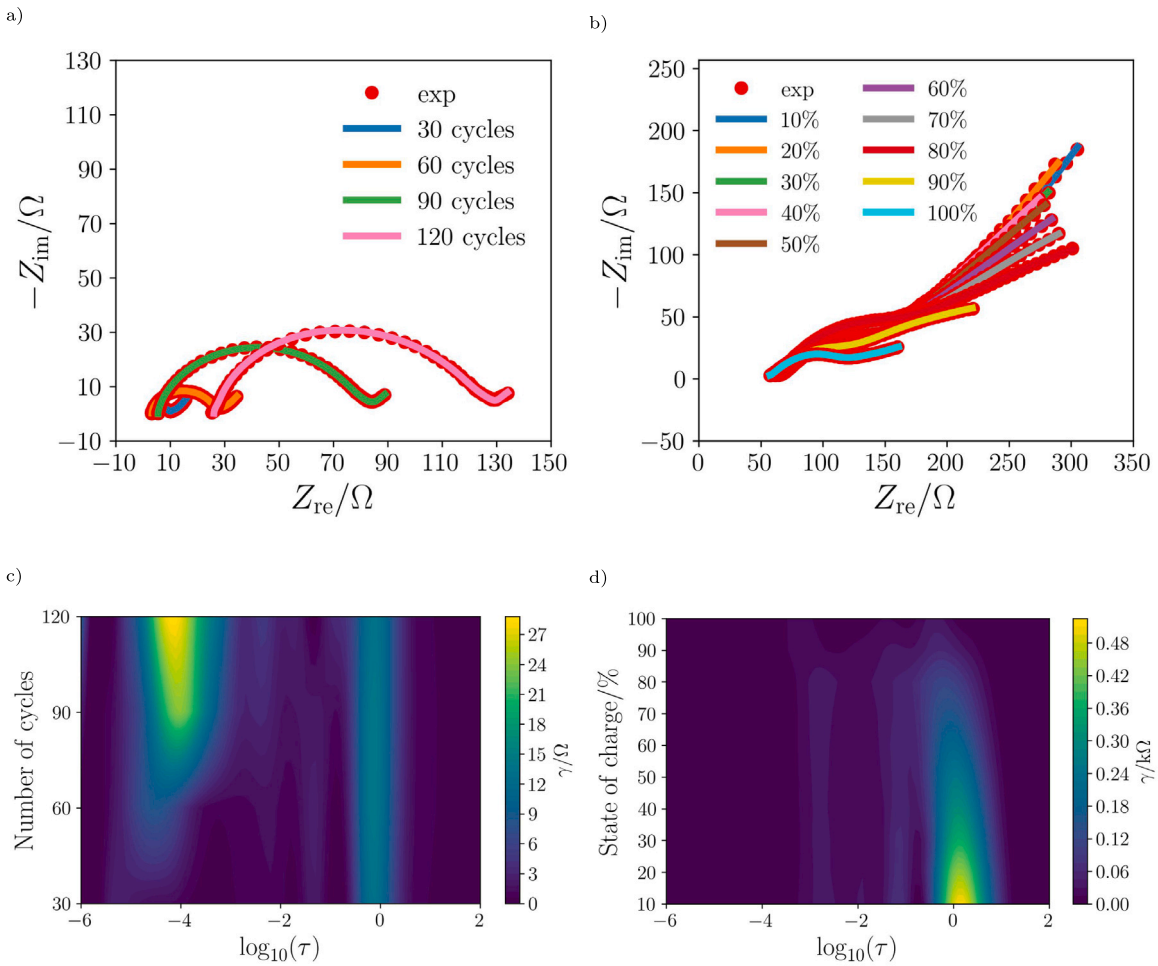

Fig. 7. For the LFP battery, (a) Nyquist plots of the experimental and DRT-fitted impedances, and (c) contour plots of the recovered DRTs. The corresponding plots for the ML621 battery are shown in panels (b) and (d).

3.3. Future directions

While the entropy method shows considerable potential for accurately recovering the DRT and enhancing DRT analysis, there are still areas of potential improvement. Initially, we demonstrated that integrating the entropy penalty with additional penalties based on the Euclidean norm (Eq. (11)) allows us to more effectively understand the impact of experimental conditions on the recovered DRTs (Sections 3.1.3 and 3.2). However, other norms, such as the 1-norm and infinity norm, could enhance DRT reconstruction by introducing sparsity [37]. For instance, lasso regression, which employs the 1-norm for the penalty to reduce the number of non-zero DRT components, recovers discrete DRTs, *e.g.*, the DRT of the resistor-capacitor association, more accurately than ridge regression [38]. Second, and regarding the regularization matrices $\mathbf{L}_\tau$ and $\mathbf{L}_\psi$, exploration of the differentiation order in (11) could address scenarios where the DRT is discontinuous concerning the timescales [39]. Boundary conditions could also be implemented when defining these two matrices [40,41]. Third, the three regularization parameters λ, λ_τ, and λ_ψ in (11) were determined empirically for real data (discussed in Section 3.2), but they could also be optimized using the multidimensional L-curve approach (Section 3.1.2).

4. Conclusions

This study advances DRT deconvolution by introducing entropy-based regularization as an alternative to conventional matrix differentiation penalties. The method improves DRT reconstruction accuracy while effectively suppressing spurious peaks that often compromise standard regularization techniques. Performance remains consistent across impedance models and experimental conditions, including scenarios with high noise levels, and it enables uncertainty quantification through DRT sampling. Moreover, the entropy-based method's extension to multiple experiments enables concurrent analysis of impedance spectra measured across different experimental conditions, providing insights into how electrochemical processes evolve with operational parameters. Analysis of experimental data from batteries, fuel cells, and solar cells demonstrates that the entropy-based approach yields interpretable relaxation time distributions while maintaining accurate impedance regressions. This capability in handling multidimensional datasets facilitates the investigation of electrochemical process dependencies on operational parameters. In short, entropy-based regularization offers a valuable tool for DRT deconvolution, combining improved accuracy with practical applicability across electrochemical systems. This advancement enables more detailed characterization of electrochemical processes through impedance analysis, contributing to both fundamental research and applications in energy conversion and storage technologies.

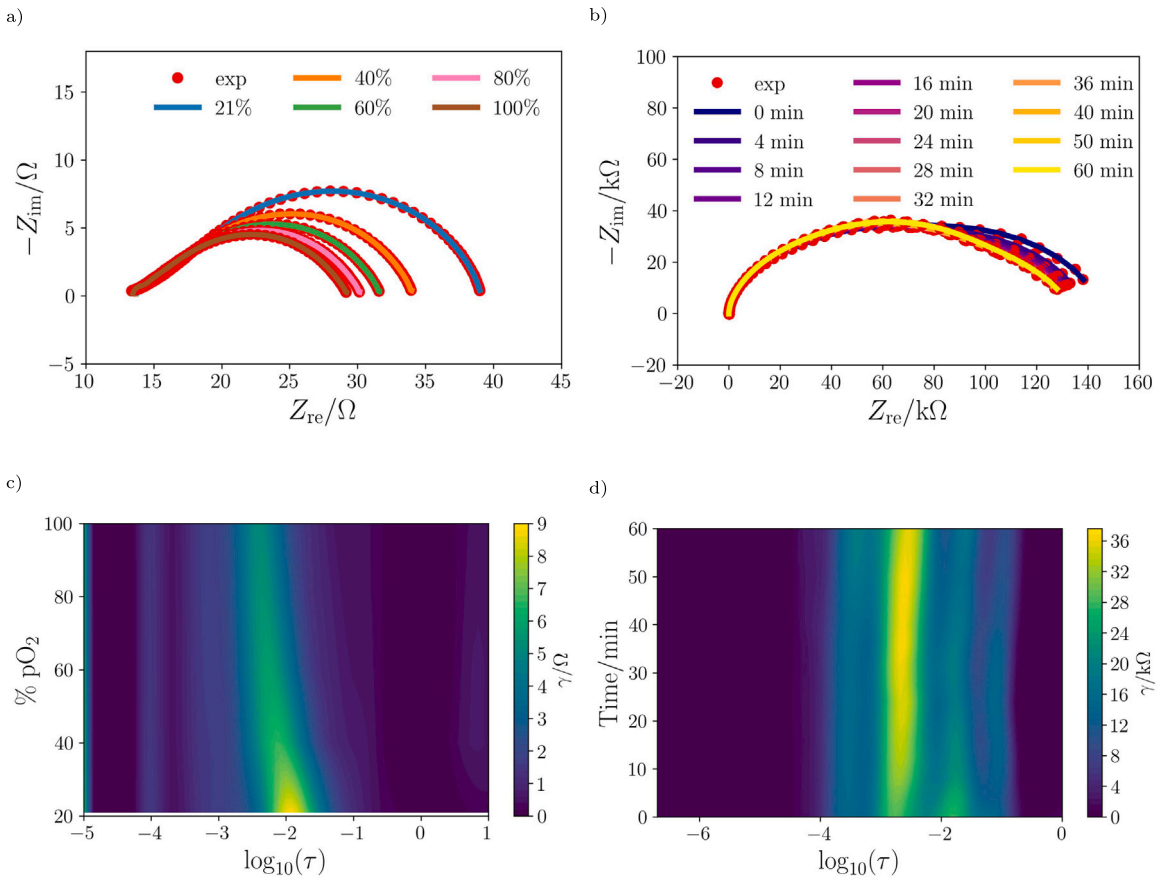

Fig. 8. For the protonic ceramic fuel cell, (a) Nyquist plots of the experimental and DRT-fitted impedances, and (c) contour plots of the recovered DRTs. The corresponding plots for the perovskite solar cell are shown in panels (b) and (d).

CRediT authorship contribution statement

Baptiste Py: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Zilong Wang:** Writing – review & editing, Visualization, Conceptualization. **Yuhao Wang:** Writing – review & editing, Visualization, Conceptualization. **Francesco Ciucci:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used Claude (Anthropic) and Gemini (Google) to improve the language and clarity of certain parts. After using these tools, the authors reviewed and edited the content. Dr. Mark Ellwood carried out further editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jpowsour.2025.236910.

Data availability

Data will be made available on request.

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